

Flood Prediction Using Machine Learning: Integrating Classification and Regression Approaches

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Abstract: Floods pose great risks to human life, infrastructure, and the environment. The application of machine learning techniques in predicting the occurrence of floods can enhance preparedness for disasters and mitigation strategies. This present study employs a dataset containing 50,000 samples and 21 features covering environmental, infrastructural, and socio-economic factors in developing predictive models. Different regression and classification techniques were applied, such as XGBoost, Random Forest, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN). Results indicate that XGBoost yielded the highest regression accuracy ($R^2 = 0.925$), with SVM and Logistic Regression being superior to other classifiers. Results attest to the effectiveness of machine learning for flood prediction and its adaptability to different regional contexts.

Keywords: flood prediction, machine learning, regression, classification

I. INTRODUCTION

Flooding is described as one of the most destructive natural disasters, causing immense losses to human settlements, infrastructure systems, and natural environments. Rising frequency and severity of floods because of climate change, deforestation, and rapid urbanization indicate the urgent need for improved flood forecasting models. Conventional hydrological models founded on physical and statistical frameworks tend to have difficulties in capturing the intricate and evolving nature of environmental variables that govern flood risk. Machine learning has emerged as a powerful tool for flood prediction, offering data-driven approaches with the ability to explain intricate interdependencies between different variables while improving the accuracy of predictions.

This study aims to use machine learning algorithms for the prediction of the likelihood of flooding and classify the ranges of flood vulnerability by environmental, infrastructure, and socio-economic status.

With a dataset comprising 50,000 samples and 21 distinctive features, this research explores the predictive

performance of a range of regression and classification models, including XGBoost, Random Forest, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN). The dataset incorporates principal drivers such as monsoon intensity, topographic drainage, deforestation, urbanization, and the implications of climate change, hence bringing a holistic view regarding the determinants of flood risk.

The research parameters are both classification and regression problems. The regression models predict continuous values of flood likelihood, and classification models partition flood risk levels into ranges of Low, Medium, and High risk. The research also takes into consideration the models' robustness in varying geographical locations to study the generalization capacity of the suggested approach. To achieve these goals, the study addresses the following research questions:

(1) How accurately can machine learning algorithms predict flood probability based on environmental and socio-economic factors?

(2) Which machine learning algorithm provides the best performance for flood probability regression?

(3) How accurately can classification models predict flood risks into the Low, Medium, and High categories?

(4) Can machine learning models be transferred across regional contexts for flood prediction?

Responding to these questions is essential to the formulation of effective and dependable flood forecasting systems that will help policymakers and disaster management officials make informed decisions.

The main objectives of this research are:

(1) To model and evaluate machine learning algorithms to predict flood probability using a properly organized dataset,

(2) To compare various regression techniques against performance metrics such as R^2 , RMSE, and MAE,

(3) To implement classification algorithms to categorize flood hazards and evaluate their performance based on precision, recall, and F1-score

(4) To explore the potential of machine learning models for flood prediction in varied geographical areas.

Through the application of ultramodern machine learning algorithms, the accuracy of flood prediction is enhanced in this study and a scalable system for risk analysis is provided. The outcomes are intended to inform disaster preparedness initiatives for the authorities to implement proactive solutions in mitigating the effects of floods. As uncertainty associated with extreme weather phenomena increases, incorporating machine learning into flood prediction models is a remarkable achievement in constructing resilient communities and streamlining disaster response initiatives.

II. LITERATURE REVIEW

Recent studies have proven the effectiveness of ML techniques in flood forecasting. For instance, Dimri et al. [1] developed an ML-based flood prediction system using XGBoost, CatBoost, and Random Forest, achieving over 90% accuracy. Their study highlighted the superiority of gradient boosting algorithms in flood forecasting but showed the need for further research on localized flood events. Similarly, Wang et al. [2] employed a spatio-temporal deep learning model integrating Temporal Convolutional Networks (TCN), Gated Recurrent Units (GRU), and Random Forest for streamflow prediction. Their approach significantly outperformed baseline models and demonstrated the effectiveness of probabilistic and lead-time forecasting, though it was limited to a single geographical region.

Several studies have explored ensemble learning and decision tree-based models for flood prediction. Gan et al. [3] investigated the application of LightGBM in predicting estuarine water levels based on historical water level data from the Yangtze Estuary. Their model achieved a Root Mean Square Error (RMSE) of 0.12–0.17 m, indicating strong predictive capability for short-term forecasts. However, the study emphasized the need for incorporating multi-source input data to enhance prediction accuracy. In another study, Serrano-López et al. [4] developed ML models, including K-Nearest Neighbors (KNN), Random Forest, XGBoost, and Linear Regression, to predict flooding in the Ter River. Their findings showed that ML models could provide valuable insights for short-term flood prediction, yet long-term forecasting models remained underexplored.

ML techniques have also been widely used in flood susceptibility mapping. Demissie et al. [5] applied Logistic Regression, Random Forest, SVM, KNN, AdaBoost, and XGBoost to assess flood susceptibility using 13 flood predictor variables. Their study achieved accuracies ranging from 0.82 to 0.97, with XGBoost and Random Forest emerging as the top-performing models. However, they noted the need for multi-resolution approaches to improve the scalability of flood prediction models. Li et al. [6] further advanced flood prediction research by integrating LSTM, GRU, and Random Forest for monthly runoff prediction. Their results indicated a significant improvement over traditional univariate models, reinforcing the effectiveness of multi-variable ML approaches. Despite these advancements, more

research is needed to improve predictive accuracy for extreme weather events.

Rainfall-runoff modeling is another area where ML techniques have been successfully applied. Singh et al. [7] compared Multiple Linear Regression (MLR), Multivariate Adaptive Regression Splines (MARS), SVM, and Random Forest for runoff prediction in the Gola watershed. Their study found that Random Forest achieved the highest accuracy ($R^2 = 0.96$, $NSE = 0.94$). However, the applicability of their model was restricted to smaller watersheds, needing further studies on larger regions. Ahmed and Lee [8] extended ML applications to urban flooding and public transit systems, using Random Forest, XGBoost, KNN, and Naïve Bayes to assess flood risks. Their model achieved over 95% accuracy, highlighting ML's potential for urban flood risk assessment, but real-time flood prediction systems remain an area for improvement.

Deep learning models have also gained traction in flood prediction. Hosseiny [9] introduced a U-Net-based deep learning model for river flooding prediction, which demonstrated superior performance over traditional hydrological models. Their study emphasized the need for larger-scale datasets to enhance model generalizability. Finally, Kundra et al. [10] explored Multi-Layer Perceptron (MLP), CatBoost, and Extra-Tree classifiers for early flood prediction using rainfall data, achieving 98.34% accuracy with CatBoost. While their model demonstrated high predictive accuracy, incorporating additional predictive factors such as topographical and infrastructural data could further improve flood forecasting.

III. RESEARCH METHODOLOGY

The methodology for this study involves a systematic approach to predicting flood probability using machine learning techniques. The dataset used contains 50,000 rows and 21 numeric features, all of which are suitable for regression and classification tasks without requiring extensive preprocessing. The methodology includes several key steps: data preprocessing, exploratory data analysis (EDA), feature selection, model training, hyperparameter tuning, evaluation using performance metrics, and result visualization.

The dataset was first examined for missing values, duplicate rows, and inconsistencies using the `.info()`, `.describe()`, and `.duplicated().sum()` functions in pandas. The data was confirmed to be complete, with no missing or duplicate values. Histograms and box plots were generated to analyze the distribution and presence of outliers, with findings indicating that most features followed a near-normal distribution and contained minimal extreme values. A correlation matrix and heatmap were used to identify the relationships between independent variables and the target variable (*FloodProbability*). The highest correlated features included topography drainage, deteriorating infrastructure, river management, and watersheds. Feature importance was further analyzed using a Random Forest model, which reinforced the significance of these variables in flood prediction.

Following the preprocessing phase, the dataset was split into 80% training and 20% testing subsets to allow for model evaluation. Since all features were numeric, the StandardScaler was applied to standardize the data, ensuring that all variables had a mean of zero and unit variance, which was particularly necessary for models sensitive to scale, such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN). The modeling phase included the implementation of multiple regression models to predict the continuous target variable, FloodProbability. The models tested were Linear Regression, XGBoost Regressor, Random Forest Regressor, KNN Regressor, and Support Vector Regressor (SVR). Performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) were used to evaluate model accuracy. Hyperparameter tuning was conducted for the KNN model using GridSearchCV, optimizing parameters such as the number of neighbors, distance metric, and weight function. The best-performing model was selected based on its ability to minimize RMSE and maximize R^2 .

Additionally, classification models were implemented to categorize flood risk levels into three distinct classes: Low Risk (FloodProbability < 0.4), Moderate Risk ($0.4 \leq$ FloodProbability < 0.7), and High Risk (FloodProbability $\geq$ 0.7). The dataset was analyzed for class distribution, revealing an imbalance, with the majority of instances falling within the Moderate Risk category. The classification models tested included Logistic Regression, Random Forest Classifier, Support Vector Machine (SVM), XGBoost Classifier, and K-Nearest Neighbors (KNN) Classifier. These models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. To further improve model performance, hyperparameter tuning was applied to SVM using GridSearchCV, optimizing parameters such as kernel type, regularization (C), and gamma. This step ensured that the model generalized well to unseen data and was not overfitting to the training set. The final models were compared based on their predictive performance and classification accuracy, with visualizations such as scatter plots and confusion matrices used to interpret their outputs. Throughout the process, cross-validation was employed to validate model robustness and prevent overfitting. The methodology followed in this study ensures a rigorous and systematic approach to flood prediction using machine learning techniques, with an emphasis on data quality, feature importance analysis, model selection, hyperparameter tuning, and comprehensive evaluation using appropriate performance metrics. The techniques implemented provide a scalable and adaptable framework for flood prediction, which can be applied to different geographical regions with minimal modifications.

IV. RESULTS

The results of this study show the effectiveness of various machine learning models in predicting flood probability and classifying flood risk levels. The regression models were evaluated based on key performance metrics, including Mean Absolute Error

(MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2). The classification models were assessed using precision, recall, F1-score, and accuracy. The regression analysis results, shown in TABLE I, reveal that XGBoost Regressor achieved the best performance with an R-squared value of 0.9252 and the lowest RMSE of 0.0136, indicating a strong predictive ability. Linear Regression, despite achieving an R^2 of 1.0, likely overfitted the data due to its unrealistic performance metrics. The Random Forest Regressor, Support Vector Regressor (SVR), and K-Nearest Neighbors (KNN) Regressor provided competitive results, with R^2 values of 0.7305, 0.7203, and 0.7747, respectively, but fell short of XGBoost in terms of predictive accuracy.

TABLE I. REGRESSION RESULT

Model	MAE	MSE	RMSE	R-squared
Linear Regression	8.155143e-17	1.152384e-32	1.073492e-16	1.000000
Random Forest Regressor	2.046965e-02	6.710998e-04	2.590559e-02	0.730545
XGBoost Regressor	1.081419e-02	1.862741e-04	1.364823e-02	0.925209
Support Vector Regressor (SVR)	2.015037e-02	6.967011e-04	2.639510e-02	0.720266
K-Nearest Neighbors Regressor	1.859030e-02	5.612321e-04	2.369034e-02	0.774659

For classification tasks, TABLE II presents the performance of different models. The Logistic Regression model demonstrated near-perfect performance, achieving an accuracy of 99.99%, followed closely by the Support Vector Machine (SVM) model with an accuracy of 99.57%. XGBoost and Random Forest classifiers also performed well, with XGBoost reaching 98.69% accuracy and Random Forest achieving 98.22%. KNN, while still effective, had slightly lower performance metrics, achieving 98.44% accuracy.

TABLE II. CLASSIFICATION RESULT

Model	Precision	Recall	F1 Score	Accuracy
Logistic Regression	0.999800	0.9999	0.999850	0.9999
Random Forest	0.964717	0.9822	0.973380	0.9822
XGBoost	0.986070	0.9869	0.983473	0.9869
SVM	0.995619	0.9957	0.995372	0.9957
KNN	0.984544	0.9844	0.978375	0.9844

These results suggest that the selected models are highly effective in predicting flood probability and classifying flood risk levels based on environmental and socio-economic factors. However, the overfitting

observed in Linear Regression indicates that further refinement may be necessary to improve generalizability. Future work should explore additional feature engineering techniques, ensemble learning strategies, and real-time data integration to enhance the model's predictive capabilities. Furthermore, expanding the dataset to include different geographic regions and incorporating climate projections could improve the adaptability and robustness of the model. Fine-tuning hyperparameters through more advanced optimization techniques, such as Bayesian optimization, could further enhance model performance. Lastly, deploying the model in real-world flood risk management systems and validating its predictions with real-time flooding events will be critical for assessing its practical applicability and impact.

V. CONCLUSION

The results of this study highlight the effectiveness of various machine learning models in predicting flood probability and classifying flood risk levels. The regression analysis demonstrated that the XGBoost Regressor outperformed other models with an R-squared value of 0.9252 and the lowest RMSE of 0.0136, indicating a strong predictive ability. However, the Linear Regression model exhibited overfitting, achieving an R^2 of 1.0 with unrealistic performance metrics, suggesting that it may not generalize well to unseen data. The Random Forest Regressor, Support Vector Regressor (SVR), and K-Nearest Neighbors (KNN) Regressor produced competitive results but did not surpass XGBoost in predictive accuracy. For classification tasks, the Logistic Regression model achieved near-perfect performance with an accuracy of 99.99%, making it the most effective model for flood risk classification. The Support Vector Machine (SVM) also performed exceptionally well with 99.57% accuracy, while XGBoost and Random Forest classifiers followed closely with accuracies of 98.69% and 98.22%, respectively. Although KNN demonstrated strong classification performance (98.44% accuracy), it was slightly lower than the other models.

These findings confirm that machine learning techniques can successfully analyze flood-related data and predict flood probability with high accuracy. However, certain challenges remain. The overfitting observed in Linear Regression suggests a need for further refinement and regularization techniques to improve generalizability. Additionally, the classification models, although highly accurate, should be tested on more diverse datasets to ensure robustness across different geographical and environmental conditions. To enhance the predictive power and applicability of these models, future work should explore feature engineering and data expansion by incorporating more environmental,

hydrological, and socio-economic variables, as well as expanding the dataset to include different geographic regions and climate projections to improve adaptability and robustness. Further tuning hyperparameters using advanced techniques like Bayesian optimization and implementing ensemble methods could enhance model performance. Real-time data integration, including satellite data, IoT sensor readings, and real-time weather reports, will enable dynamic flood prediction models.

Deploying the models in practical flood risk management frameworks and validating predictions with real-world flooding events will be crucial for assessing real-life applicability. In conclusion, this study confirms that machine learning models are highly effective in flood prediction and classification, with XGBoost emerging as the most reliable regression model and Logistic Regression excelling in classification tasks. By refining these models and expanding their datasets, they can serve as valuable tools for early warning systems and disaster preparedness, ultimately helping mitigate the impact of floods and improving community resilience.

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